University of Science and Technology of Hanoi

ICT Department

Bachelor Thesis Report

Low-Entropy Attacks on 6-Digit PINs: Modeling Guessability under Personal-Information Bias

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by

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ABSTRACT

This thesis investigates the security of human-chosen 6-digit PINs under low-entropy attacks. Although a 6-digit PIN has a theoretical space of one million combinations, user behavior introduces strong biases that significantly reduce practical security.

This work models human-chosen PIN distributions and evaluates multiple attack strategies, including random guessing, frequency-based guessing, and personal-information-assisted attacks. The study focuses on limited-attempt scenarios such as top-1, top-3, top-5, and top-10 guesses.

Experimental results demonstrate that user bias and personal information leakage greatly increase attack success rates compared to the uniform random model. The thesis highlights the importance of guessability-based metrics and provides recommendations to improve PIN security.

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Chapter 1

Introduction

1.1 Background

Personal Identification Numbers (PINs) are widely used for authentication in various systems, including banking, mobile devices, and online services. A 6-digit PIN provides a theoretical space of one million possible combinations. However, in practice, users tend to select PINs based on memorable patterns, such as dates, repeated digits, and sequences.

1.2 Problem Statement

The practical security of a PIN depends not only on its theoretical key space but also on how users choose their PINs. Human bias reduces the effective entropy of the system, making it vulnerable to guessing attacks. Furthermore, personal information such as birth dates can further increase the success rate of targeted attacks.

1.3 Objectives

The objective of this thesis is to evaluate the vulnerability of human-chosen 6-digit PINs under low-entropy attacks. The study aims to model user behavior, simulate attack strategies, and measure security using guessability-based metrics.

1.4 Contributions

This thesis provides:

- A probabilistic model for human-chosen PIN distributions
- Simulation of multiple attack strategies
- Evaluation of security under limited guessing attempts
- Analysis of the impact of personal information leakage

1.5 Structure of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews related work on PIN security and low-entropy attacks. Chapter 3 presents the methodology. Chapter 4 discusses experimental results. Finally, Chapter 5 concludes the thesis.